

GAYATRI VIDYA PARISHAD COLLEGE FOR DEGREE & PG COURSES (A)

Evaluation Answer Key Blueprint

Course: Data Structures Using C (CS1203) Examination: MID-I Max. Marks: 30

Q1(a) Linear vs Non-Linear Data Structures

Expected Answer / Key Points	Marks
Definition of Linear Data Structure	0.5
Definition of Non-Linear Data Structure	0.5
One suitable example for each	1

Q1(b) Priority Queue

Expected Answer / Key Points	Marks
Correct definition	1
Any two valid applications (0.5 each)	1

Q1(c) Threaded Binary Tree

Expected Answer / Key Points	Marks
Definition	1
Purpose/advantage of threads	1

Q2(a) Infix to Prefix Conversion

Expected Answer / Key Points	Marks
Algorithm/steps	3
Correct use of stack / logical procedure	1
Illustrative example with correct prefix expression	2

Q2(b) Reverse Singly Linked List

Expected Answer / Key Points	Marks
Variable initialization	1
Loop and traversal	1
Correct pointer modification	2
Head update and termination	1

Correct pseudocode/algorithm presentation	1
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Q3(a) Circular Queue using Arrays

Expected Answer / Key Points	Marks
Representation and initialization	2
Enqueue algorithm	2
Dequeue algorithm	2

Q3(b) Arrays vs Linked Lists

Expected Answer / Key Points	Marks
Memory allocation comparison	2
Access time comparison	2
Insertion/Deletion comparison	2

Q4(a) Tree Traversals

Expected Answer / Key Points	Marks
Correct tree construction/example	2
Inorder traversal	1.5
Preorder traversal	1.5
Postorder traversal	1

Q4(b) Applications of Trees

Expected Answer / Key Points	Marks
Any six valid applications (1 mark each) OR four well-explained applications (1.5 marks each)	6

Q5(a) Binary Search Tree

Expected Answer / Key Points	Marks
Correct BST construction	3
Correct deletion of node 30	3

Q5(b) Heterogeneous Binary Trees

Expected Answer / Key Points	Marks
Definition	2

Structure/diagram	2
One practical application	2